Artificial Intelligence and its implications on Water Conservation and Consumption

Nowadays, we are hearing about the growth and the utilization of Artificial Intelligence increasing in each and every field. Regularly, we read that someone created something innovative and revolutionary. Every day the technical field is advancing further and Artificial Intelligence has a great impact on research and development of many useful and innovative things. But have you thought about how Artificial Intelligence can be useful in our day to day lives and how we can preserve resources which we need to live? How can we use Artificial Intelligence for Water Conservation and Consumption?

To research about a water body we need a lot of data. However, to collect this data we need to give a huge amount of time and resources. It is hard to constantly observe and detect any changes occurring in the water body as we can only observe for a sample but it is impossible to look for the entire water body. To solve this problem we can use AI which can help us predict the outcomes and help us calculate complex problems using the dataset currently present for any water body.

To calculate the Water Quality Index(WQI) we need to check for multiple parameters and the process is laborious and has several limitations which can cause inaccuracy in determining the WQI (*Artificial Intelligence in Water Quality Monitoring: A Review of Water Quality Assessment Applications*, 2025). The recent developments in AI tools for assessing water quality has the potential to provide significant improvements. It may reduce the computation power, efforts, cost and increase the accuracy of WQI as it can calculate and compute multiple parameters easily. In the future the combination of AI and remote sensing can help and change the methods of collecting data using satellite data and on-site sensor data. For instance, in a study carried out on monitoring data from Thailand for the period 2016–2021, it has been possible to reduce the number of parameters for WQI calculation from 13 to four, using ANN and the Bootstrap method (Chawishborwornworng, 2024).

Artificial Intelligence technology has potential to revolutionize the future of water and wastewater systems. AI can solve problems across the water sector like water quality monitoring by analyzing data from sensors placed in water bodies. AI algorithms can detect changes in real time to check for quality and identify potential contaminants or public health hazards such as pollution plumes and waterborne pathogens. AI can also be used to detect these problems and give us solutions to solve these problems easily.

We know that these days conserving and preventing water wastage is most important so we have to minimize water wastage in all ways. AI can help identify and prevent leaks in water distribution systems. By analyzing data from sensors and meters. AI algorithms analyze the data from leak data and can detect anomalies in water flow and pressure indicating possible leaks. It can also check for areas in the pipeline where leaks can occur. This approach can save a significant amount of water and reduce the cost associated with leaks.

Our main goal is to preserve and conserve water. This can be done with the help of AI as it can help optimize water usage by analyzing data from various sources, such as weather patterns, soil moisture levels, and crop water requirements. AI can utilize this data to get insights and to make decisions on water allocation for farming and also help construct infrastructure accordingly. Also not just farming, it can help us calculate our daily usage of water in our homes. During the water foundation course in Ahmedabad University there is an exercise in which there is an app which helps you calculate your daily usage of water in your house. Websites like:

- https://watercalculator.org/wfc2/q/household/
- https://askfilo.com/user-question-answers-smart-solutions/how-much-water-is-used-daily-in-your-home-calculate-the-3335353937363138

So, these websites help us observe and calculate our water footprint and how much water we waste, which can help us optimize our usage of water to save more water.

Artificial Intelligence technology can also assist in managing water resources more efficiently by analyzing data on water availability, usage patterns and population growth. Using AI algorithms, the government authorities can get insights from the data and take informed decisions on water allocation to different areas. When one of the water body resources is getting dried up due to daily usage of water, government authorities can start preserving these bodies and set limits to water usage.
In 2020, the city of Tucson, Arizona implemented AI technology in an effort to be more proactive in managing its water system, consisting of over 4,600 miles of distribution water main pipes (Bonney, n.d.).

In practise four examples of the use of AI, machine learning and LLMs in the water and sanitation sectors which use the following AI tools:

1. **Safe Water Optimization Tool**
2. **Faecal Sludge Snap App**
3. **WASH AI**
4. **Smartphone App for Identifying Poor Rural Households in Ghana**

These tools are used in different sections but are not an exhaustive list. “…deep learning has the potential to solve challenges by filling in spatial and temporal data through training to improve its predictions about surface water” (*AI in Water and Waste Water Sectors*, 2025).

Until this point we have discussed the application of Artificial Intelligence Technologies which can be beneficial in water conservation, consumption monitoring and resource management. However, establishing these industries and operation of these AI technologies need a significant amount of land space, infrastructure and natural resources. There are consequences of building data centers for AI on natural resources and has a significant impact on water bodies too.

Data centers and AI need clean and treated water to avoid blockages and the growth of bacteria in pipes. In the case of Google's self owned datacenters, only 20 percent of withdrawn water is discharged back to the water treatment plants with the rest of water lost to evaporation. AI and its data centers require freshwater to cool servers and produce the electricity that powers them. In 2022, Google, Microsoft, and Meta used an estimated 580 billion gallons of water to provide power and cooling to data centers and AI servers. That's enough water to meet the annual needs of 15 million households (Murray, 2025).

We saw above that there are direct impacts of AI technologies on water. But along with that there are indirect consequences too. AI technologies depend on powerful hardware such as GPUs, servers, semiconductors, and electronic components. When these devices become outdated, they generate electronic waste like lead, mercury and cadmium which are toxic and if not disposed of properly, this waste can contaminate rivers, groundwaters and soil. The construction of large AI data centers require land cleaning, large infrastructure and urban development and alteration of nearby water systems. This can disturb wetlands, rivers and aquatic habitats. The construction of AI data centers may damage aquatic ecosystems and wetlands. For example, a 2025 environmental report

noted that an Amazon data center project in New Carlisle would destroy approximately 10 acres of wetlands (*Measuring the Environmental Cost of Artificial Intelligence and Its Data Centers*, 2025). AI powered data centers also contribute to greenhouse gas emissions and these large emissions can disturb the water cycle and can cause defects in the seasonals changes and as we know they affect climate change.
Current estimates suggest that data centers generate approx 180 million metric tons of indirect carbon dioxide every year (*AI and Climate Change – Energy and AI – Analysis*, n.d.).

A powerful statement published in an article written by Lincoln Institute which goes like, “A low hum emerges from within a vast, dimly lit tomb, whose occupant devours energy and water with a voracious, inhuman appetite. The beige, boxy data center is a vampire of sorts—pallid, immortal, thirsty. Sheltered from sunlight, active all night. And much like a vampire, at least according to folkloric tradition, it can only enter a place if it’s been invited inside” (Gorey, 2025). The above line comprehends what all we discussed earlier about the consequences of using AI technologies.

In conclusion, Artificial Intelligence can be both a remarkable opportunity and a serious challenge to water conservation and consumption. On one hand, AI has the power to transform how we currently monitor water quality, and how we detect leaks, wastewater management and resource allocation. Through predictive analysis, remote sensing, and real-time monitoring, AI can help governments, farmers, industries and individuals to make informed decisions to reduce water wastage and improve efficiency. On the other hand, the rapid growth of AI can be destructive for water bodies and can raise environmental concerns. The enormous amount of water usage required to cool servers, the rise of electronic waste , greenhouse gases emissions and the destruction of ecosystems due to infrastructure expansion makes the AI itself to be a destroyer of the world. Therefore, the issue here is not whether AI is good or bad, but rather how responsibly humans choose to design, implement and regulate it.

Here, the balanced approach to use these AI based technologies is that governments, industries and individuals should ensure that the AI systems are powered with the help of renewable energy sources. The water lost during cooling and evaporation should be thermodynamically preserved and reused again. The government should implement policies which encourage recycling of electronic waste, protection of wetlands and ecosystems and transparency regarding the environmental impact of these AI technologies.

Ultimately, Artificial Intelligence should not replace human responsibility but they should help humans to strengthen it and help them do their work easily. AI can become a valuable resource by securing water resources for future generations. So afterall, it depends on humanity's ability to maintain a balance between innovation and environmental responsibility which will help us in water conservation, consumption and management.

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